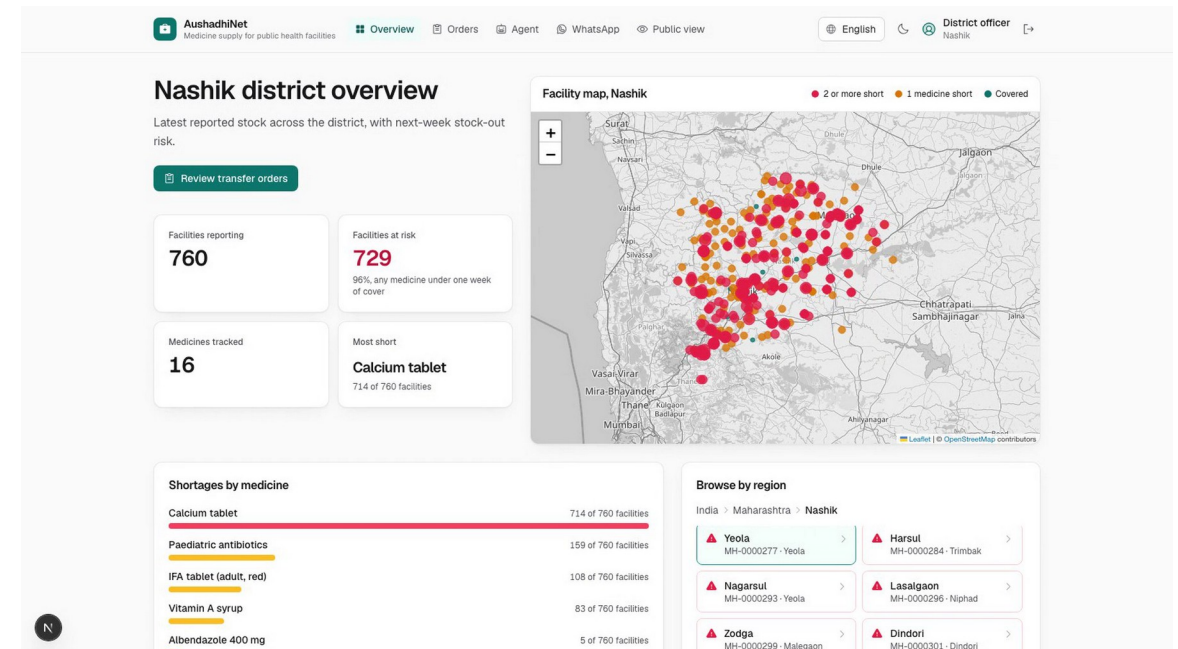


AushadhiNet

Early warning and rebalancing for public-health medicine supply, reported over WhatsApp.



Live: aushadhinetwork.onrender.com

Code: github.com/akshat333-debug/AushadhiNet

Clinics run dry while neighbours hold spare stock

Primary health centres run out of ORS, zinc, iron tablets and antibiotics.

Stock is reported late, on paper, and rarely compared across facilities.

Shortage and surplus sit a few kilometres apart; nobody sees both in time.

Frontline nurses won't adopt yet another app.

Report → Forecast → Rebalance → Human approves

1 Report

Nurse sends stock on WhatsApp in 7 languages; unsure reads are confirmed first.

2 Forecast

LightGBM predicts next-week demand and stock-out probability per medicine.

3 Rebalance

Google OR-Tools proposes transfers from nearby surplus to at-risk facilities.

4 Approve

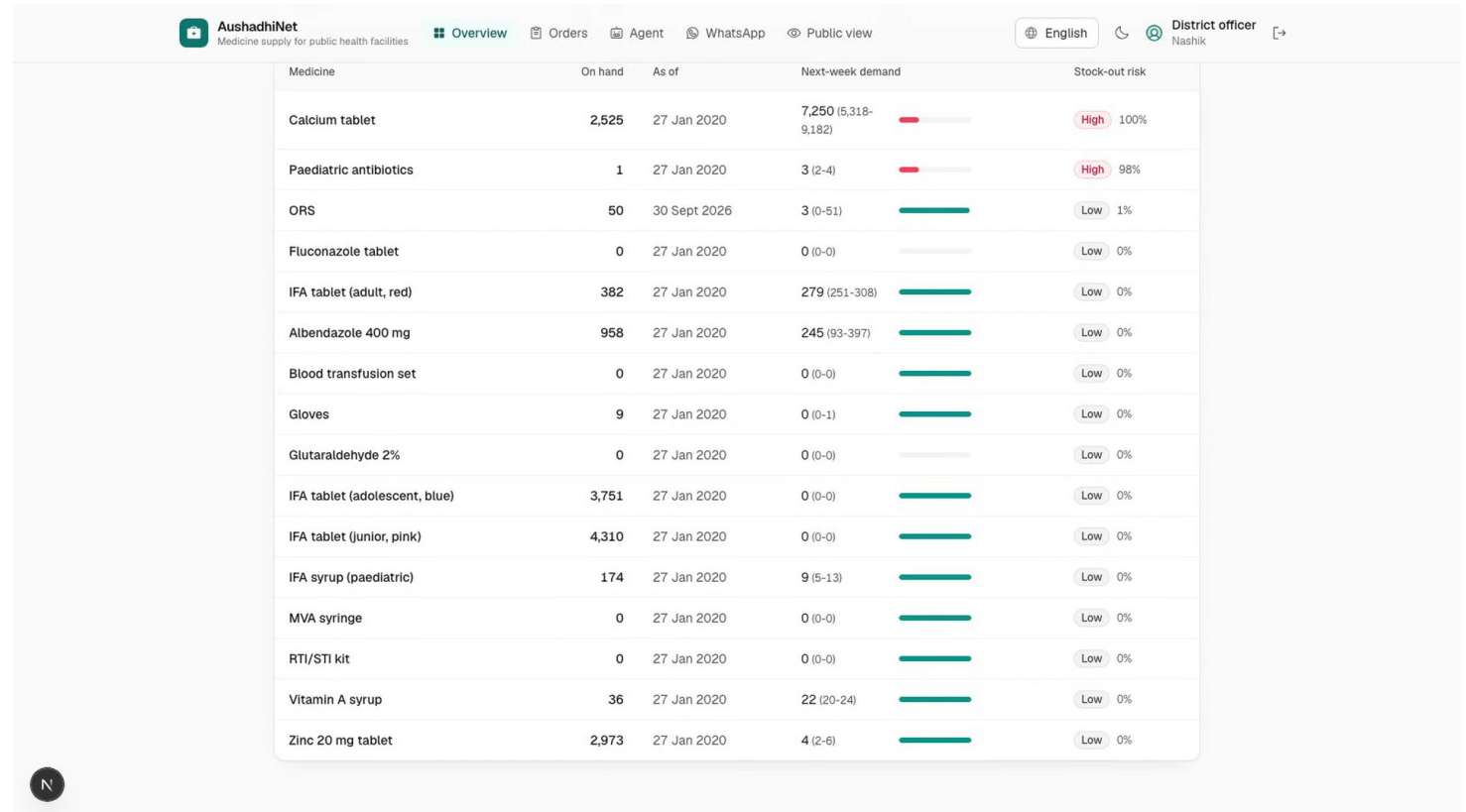
Officer approves; both facilities notified. The AI agent can explain and draft, never approve.

WhatsApp, no new app

Plain text, any script, native digits

Low-confidence reads need a Confirm tap

Nothing unconfirmed reaches the forecast



The screenshot displays the AushadhiNet web application interface. The header includes the AushadhiNet logo, the tagline "Medicine supply for public health facilities", and navigation tabs for Overview, Orders, Agent, WhatsApp, and Public view. The user is logged in as "District officer Nashik". The main content is a table with the following columns: Medicine, On hand, As of, Next-week demand, and Stock-out risk. The table lists 18 different medicines with their respective stock levels, last updated dates, demand forecasts, and stock-out risk percentages. A small circular icon with the letter 'N' is visible in the bottom left corner of the interface.

Medicine	On hand	As of	Next-week demand	Stock-out risk
Calcium tablet	2,525	27 Jan 2020	7,250 (5,318-9,182)	High 100%
Paediatric antibiotics	1	27 Jan 2020	3 (2-4)	High 98%
ORS	50	30 Sept 2026	3 (0-51)	Low 1%
Fluconazole tablet	0	27 Jan 2020	0 (0-0)	Low 0%
IFA tablet (adult, red)	382	27 Jan 2020	279 (251-308)	Low 0%
Albendazole 400 mg	958	27 Jan 2020	245 (93-397)	Low 0%
Blood transfusion set	0	27 Jan 2020	0 (0-0)	Low 0%
Gloves	9	27 Jan 2020	0 (0-1)	Low 0%
Glutaraldehyde 2%	0	27 Jan 2020	0 (0-0)	Low 0%
IFA tablet (adolescent, blue)	3,751	27 Jan 2020	0 (0-0)	Low 0%
IFA tablet (junior, pink)	4,310	27 Jan 2020	0 (0-0)	Low 0%
IFA syrup (paediatric)	174	27 Jan 2020	9 (5-13)	Low 0%
MVA syringe	0	27 Jan 2020	0 (0-0)	Low 0%
RTI/STI kit	0	27 Jan 2020	0 (0-0)	Low 0%
Vitamin A syrup	36	27 Jan 2020	22 (20-24)	Low 0%
Zinc 20 mg tablet	2,973	27 Jan 2020	4 (2-6)	Low 0%

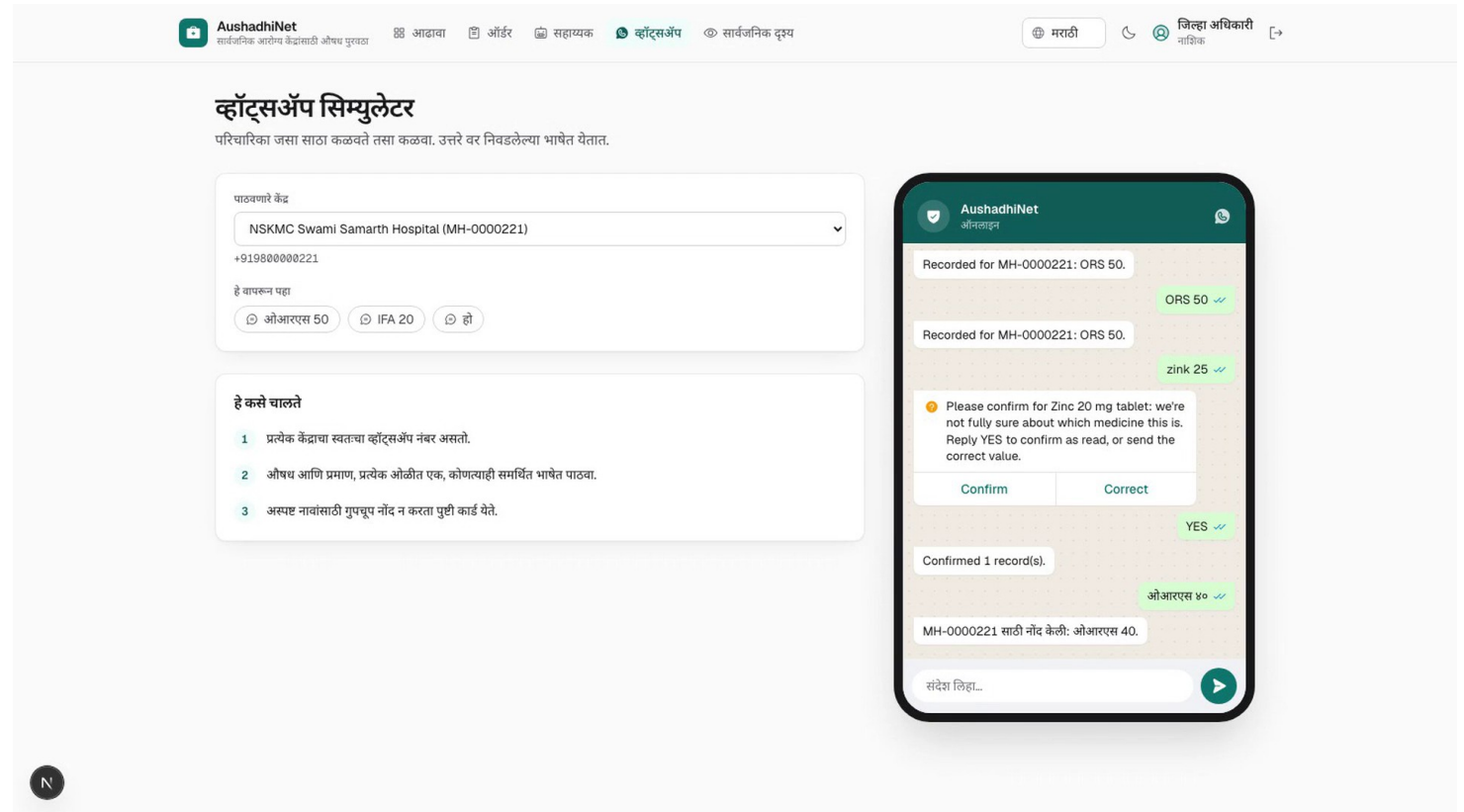
Multilingual

Seven Indian languages

English, Hindi, Marathi, Bengali, Tamil, Telugu, Kannada

UI, medicine names, replies and agent answers

Translations need native-speaker review



Transfer orders, human in the loop

Solver drafts transfers (OR-Tools)

Officer approves or rejects each

Orders are signed; both ends notified

The screenshot displays the AushadhiNet interface for a district officer in Nashik. The header includes the AushadhiNet logo, navigation links (Overview, Orders, Agent, WhatsApp, Public view), language settings (English), and user information (District officer, Nashik). The main section is titled "Transfer orders, Nashik" and includes a "Propose transfers" button. Below this, a summary states "159 drafts, 7,251 units across 186 facilities" and a "Show Drafts" dropdown. The interface shows a grid of draft transfer orders for Albendazole 400 mg. Each order card displays the drug name, quantity, source facility, destination facility, and a "min drive" time. Each card has "Reject" and "Approve" buttons. A compass icon is visible in the bottom left corner.

Drug	Quantity	Source Facility	Destination Facility	Drive Time	Status
Albendazole 400 mg	3	Umrane MH-0000317	Chinchave MH-0010793	Same location in the facility directory	Draft
Albendazole 400 mg	331	Dabhadi MH-0000331	MMC Ali Akbar Hospital MH-0000332	18 min drive	Draft
Albendazole 400 mg	104	MMC N N Wadia Hospital MH-0000333	MMC Ali Akbar Hospital MH-0000332	2 min drive	Draft
Albendazole 400 mg	615	GH Malegaon MH-0000477	MMC Ali Akbar Hospital MH-0000332	8 min drive	Draft
Albendazole 400 mg	228	GH Malegaon MH-0000477	MMC Nima I MH-0002543	5 min drive	Draft
Albendazole 400 mg	7	Kaluste MH-0002092	Kawnai MH-0000644	Same location in the facility directory	Draft
Albendazole 400 mg	15				Draft
Albendazole 400 mg	157				Draft

Explains, never acts

Answers from real forecast and stock data

Jurisdiction-scoped per officer

Refuses prompt-injection to approve orders

The screenshot displays the AushadhiNet web application interface. At the top, the header includes the AushadhiNet logo, navigation links (Overview, Orders, Agent, WhatsApp, Public view), language settings (English), and user information (District officer, Nashik). The main section is titled "Transfer orders, Nashik" and includes a sub-header "Drafts from the solver wait here. Nothing moves until an officer approves." and a "Propose transfers" button. Below this, a summary states "159 drafts, 7,251 units across 186 facilities" with a "Show Drafts" dropdown. The interface lists eight draft transfer orders for Albendazole 400 mg, each with a "Draft" label, a quantity, and a "Reject" or "Approve" button. The orders are as follows:

Quantity	From Facility	To Facility	Distance	Status
3	Umrane (MH-0000317)	Chinchave (MH-0010793)	Same location in the facility directory	Reject / Approve
331	Dabhadi (MH-0000331)	MMC Ali Akbar Hospital (MH-0000332)	18 min drive	Reject / Approve
104	MMC N N Wadia Hospital (MH-0000333)	MMC Ali Akbar Hospital (MH-0000332)	2 min drive	Reject / Approve
615	GH Malegaon (MH-0000477)	MMC Ali Akbar Hospital (MH-0000332)	8 min drive	Reject / Approve
228	GH Malegaon (MH-0000477)	MMC Nima I (MH-0002543)	5 min drive	Reject / Approve
7	Kaluste (MH-0002092)	Kawnai (MH-0000644)	Same location in the facility directory	Reject / Approve
15				
157				

Results

Measured on a frozen, held-out protocol

0.871

Forecast WAPE on real HMIS data
vs 1.078 naive (19% better)

-6.0%

Stock-out weeks in 26-week replay
(synthetic Nashik ledger)

760

Nashik facilities × 16 HMIS medicines in the pilot

7

Languages end to end

Test window frozen before tuning; first model (L2) lost to naive and stays in the run log.

Impact

AushadhiNet by the numbers

760

facilities monitored in Nashik pilot

16

essential HMIS medicines tracked

729

facilities flagged at risk before stock-out

159

transfer orders drafted by OR-Tools

7,251

units rebalanced across 186 facilities

19%

lower forecast error than naive (WAPE 0.871)

-6%

stock-out weeks in 26-week replay

7

Indian languages, end to end

362

automated tests passing (349 backend + 13 E2E)

Built on Google

Google OR-Tools · Gemini (provider seam) · Firestore / Cloud Run (scripted)

LightGBM · FastAPI · Next.js 15 · Leaflet + OpenStreetMap

348 backend tests · 13 Playwright E2E, passing on the live deployment

Thank you

Earlier warnings. Smarter sharing. People in control.

Live: aushadhinetwork.onrender.com

Next: pilot in one Nashik block with real registers.